

Planning Constructions and Justifying Them

Planning classical compass-straightedge constructions and justifications

Directions

- A *plan* is a numbered list of compass and straightedge moves precise enough that someone else could carry it out without guessing. Say where each compass point goes and whether the span changes.
- A *justification* says why the finished figure has the property claimed. Equal radii are the raw material: they give congruent segments, and congruent segments give congruent triangles.
- Measuring with a ruler or protractor is not a construction move. Neither is "draw a perpendicular" — that is the thing to be constructed.
- The problems mix planning, justifying and coordinate checking on purpose. Decide which the question is asking for before you start writing.

1. A segment $\overline{AB}$ has endpoints $A(1, 2)$ and $B(7, 10)$.

(a) Find the length of $\overline{AB}$.

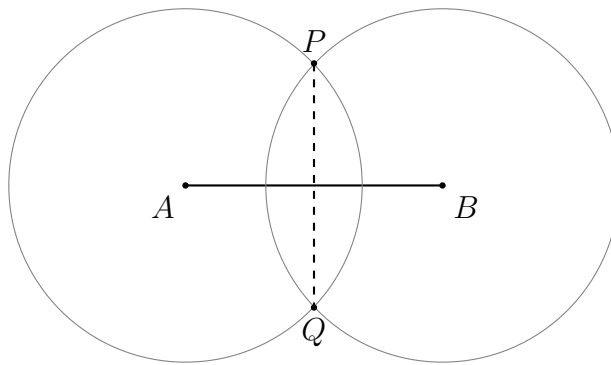
Answer: _____

(b) Write a plan for copying $\overline{AB}$ onto a ray that starts at a new point C , using only a compass and a straightedge.

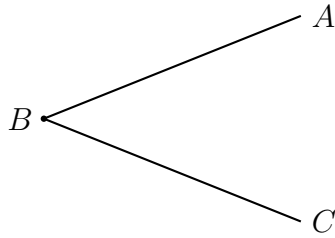
2. A construction claims that $M(1, 4)$ is the midpoint of $\overline{PQ}$, where $P(-4, 1)$ and $Q(6, 7)$. Check the claim by computing both halves. Round each length to two decimal places.

$PM = \underline{\hspace{2cm}}$ $MQ = \underline{\hspace{2cm}}$

3. To construct the perpendicular bisector of $\overline{AB}$, you set the compass wider than half of $\overline{AB}$, strike an arc centred at A and another of the *same* radius centred at B , and call the two crossings P and Q . Justify why $\overleftrightarrow{PQ}$ is the perpendicular bisector of $\overline{AB}$.



4. Write a plan for constructing the bisector of $\angle ABC$, and say what each arc guarantees about the points it creates.



5. A construction places $R(5, -1)$ so that it is claimed to be equidistant from $S(1, 2)$ and $T(9, 2)$.

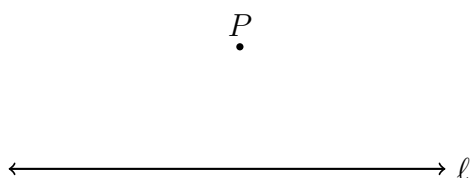
(a) Find RS . (b) Find RT .

$RS =$ _____ $RT =$ _____

(c) Explain what your two answers show about where R must lie in relation to $\overline{ST}$.

6. In the copy-an-angle construction you strike one arc across both sides of the original angle, strike an arc of the same radius across the new ray, then set the compass to the span between the two crossings on the original and carry that span, unchanged, to the new arc. Justify why the copied angle is congruent to the original, naming the congruence criterion you use.

7. Point P is not on line ℓ . Write a plan for constructing the line through P parallel to ℓ , using the copy-an-angle construction, and name the theorem that makes the two lines parallel.



8. A dilation with centre $O(0, 0)$ and scale factor 3 maps the point $A(2, 1)$ to its image A' .

(a) Find OA . (b) Find OA' . Round each to two decimal places.

$OA =$ _____ $OA' =$ _____

(c) Explain what the ratio of your two answers tells you, and what it confirms about the dilation.

9. You are given a segment $\overline{AB}$ and asked to construct an equilateral triangle with $\overline{AB}$ as one side.

(a) Write the plan, then justify why all three sides come out congruent.

(b) Suppose A is at $(0, 0)$ and B is at $(6, 0)$. The apex of the finished triangle lands at $(3, 5.196)$. Find the distance from A to the apex, to two decimal places, and check it against the side you started with.

Answer: _____

10. Synthesis. You are given a segment $\overline{AB}$ and asked to construct a square with $\overline{AB}$ as one side, reusing constructions from earlier in this worksheet.

(a) Write the plan, naming which earlier construction each stage uses.

(b) Justify why the finished figure is a square and not merely a rhombus.

(c) If A is at $(2, 1)$ and B is at $(6, 4)$, what length must every side of the finished square have?

Answer: _____